Assignment 4: Decision Tree Analysis Report

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Introduction

This report presents the implementation and evaluation of a decision tree learning algorithm using three attribute selection criteria: Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG). The analysis is based on the "adult.csv" dataset, with performance metrics computed over 20 random 80/20 train-test splits for maximum tree depths ranging from 1 to 15. The report includes plots of average accuracy versus max tree depth and observations on criterion performance, overfitting, and tree complexity.

Graphs

The following figures display the performance metrics derived from the decision tree experiments.

- **for adult.data dataset**:

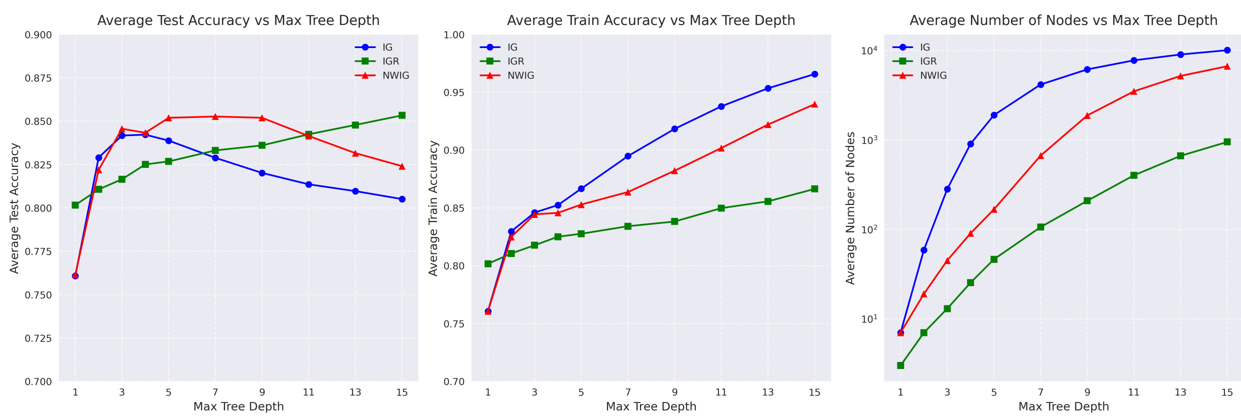

Figure 1: Performance Metrics vs. Max Tree Depth. (a) Average Test Accuracy, (b) Average Train Accuracy, (c) Average Number of Nodes.

- **for iris.csv dataset**

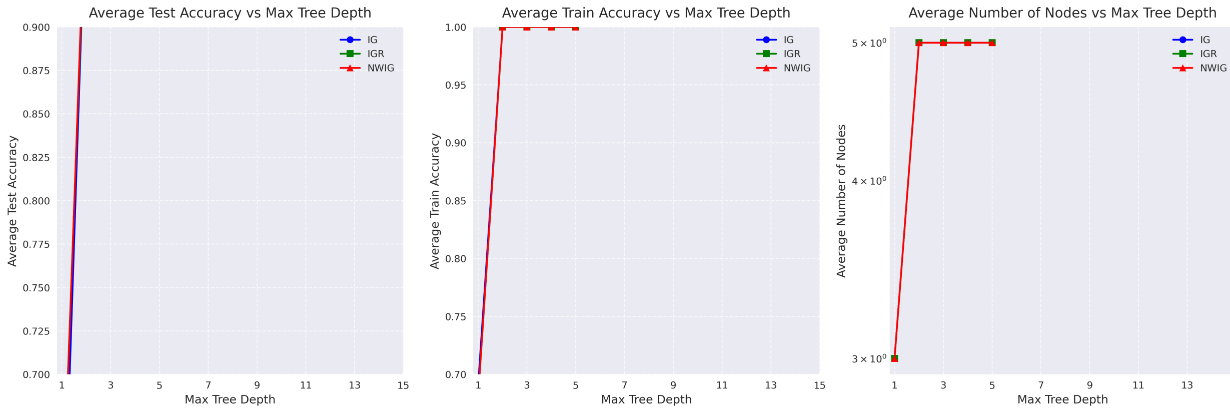

Figure 2: Performance Metrics vs. Max Tree Depth. (a) Average Test Accuracy, (b) Average Train Accuracy, (c) Average Number of Nodes.

Observations and Analysis

Analyzing the plots and data, several key insights emerge:

- **Test Accuracy Trends**: IGR demonstrates the highest average test accuracy, peaking at 0.853 at MaxDepth=15, followed by NWIG (0.824) and IG (0.805). IGR's accuracy increases steadily, suggesting better generalization across depths, while IG and NWIG peak earlier (around MaxDepth=3–5) and decline at higher depths, indicating overfitting.

- **Train Accuracy Trends**: IG achieves the highest train accuracy (0.966 at MaxDepth=15), followed by NWIG (0.940) and IGR (0.866). The rapid rise in IG and NWIG train accuracies at higher depths, contrasted with their declining test accuracies, highlights significant overfitting. IGR's train accuracy grows more gradually, aligning closely with its test accuracy, suggesting robustness.

- **Overfitting Mitigation**: IGR appears to reduce overfitting effectively, as evidenced by the smaller gap between train and test accuracies (e.g., 0.013 at MaxDepth=15) compared to IG (0.161) and NWIG (0.116). This may be due to IGR's normalization penalizing high-cardinality attributes, leading to simpler trees.

- **Tree Complexity**: IG produces the largest trees, with an average of 10,107 nodes at MaxDepth=15, followed by NWIG (6,689 nodes) and IGR (952 nodes). The logarithmic scale in the nodes plot reveals IG's exponential growth, while IGR's slower increase suggests a preference for compact structures, likely due to its intrinsic value adjustment.

- **Unexpected Patterns**: IGR's node count grows much more slowly than IG and NWIG, which could indicate that its normalization effectively balances splits, preventing over-specialization. An unexpected trade-off is that IG's high train accuracy comes at the cost of poor test performance at deeper levels, while IGR sacrifices some train accuracy for better generalization.

- **Pruning Effects**: For IG, test accuracy improves from 0.795701 (unpruned) to 0.838223 (pruned), Similarly, NWIG sees a improvement from 0.805477 to 0.851694 However, IGR shows a slight decline from 0.809428 to 0.82682 indicating that its intrinsic value normalization already mitigates overfitting to some extent, and further depth limitation may not yield as significant a benefit. Train accuracy, conversely, drops sharply with pruning—e.g., IG from 0.999918 to 0.866455 highlighting a trade-off where pruning sacrifices training performance for improved test outcomes.

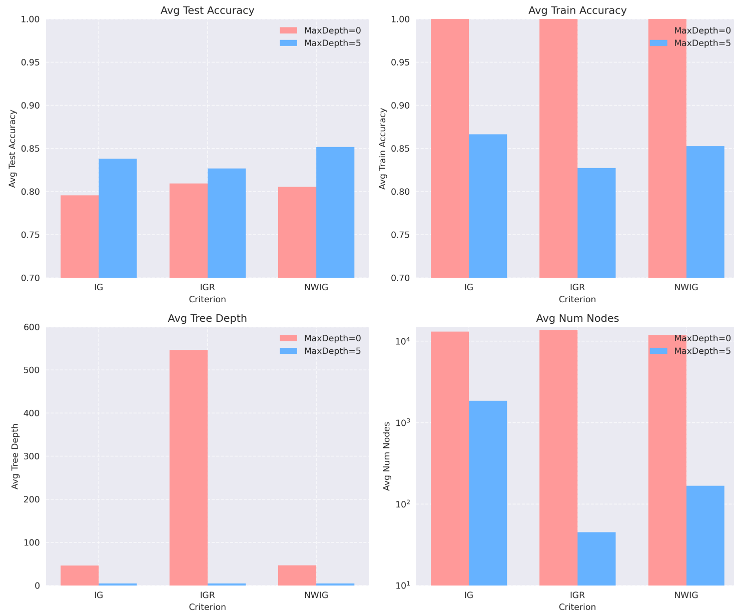

Figure 3: prun vs no pruning

- **High accuracy in iris.csv** :The "iris.csv" dataset's simplicity is evident from high accuracy at depth 0 and stable node counts up to depth 5.NWIG balances high test accuracy with simpler trees, while IG and IGR show potential overfitting at higher depths.

Conclusion

IGR outperforms IG and NWIG in test accuracy and consistency across depths, likely due to its normalization reducing overfitting. IG and NWIG build larger, more complex trees, achieving high train accuracies but suffering from overfitting at deeper levels. Future work should include no-pruning data and additional datasets to validate these findings.